

SPEAR vs. Literature-Graded Ground-Truth L–R Pairs

A benchmark against CellPhoneDB, NATMI, Connectome, log2FC, SingleCellSignalR, LIANA, and COMMOT

SPEAR project – spear-v3

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Abstract

An independent literature review graded eight candidate ligand–receptor (L–R) interaction claims (ten individual gene pairs) across the two SPEAR-analyzed public datasets for which pair-specific experimental evidence exists: mouse hypothalamus (MERFISH) and mouse embryo (seqFISH). Per instruction, the top literature grade (originally “A–”) is reported here as **Grade A** and treated as the intended ground truth: **FGF8–FGFR1** and the **BMP4–BMPRI1A/BMPRI1B** axis. We attempted to benchmark SPEAR’s own fused score, and six established expression-based CCC tools (CellPhoneDB, NATMI, Connectome, log2FC, SingleCellSignalR, and LIANA’s rank aggregate, all run via `liana-py`) plus COMMOT, against this graded set, reusing the project’s actual trained models, cached ESMC embeddings, and spatial-permutation code rather than approximating any score. The headline result is negative and, we think, more informative than a fabricated ranking would be: **every Grade A pair involves at least one gene that was never measured** in the targeted MERFISH (156-gene) or seqFISH (351-gene) panel used in this project, so *no method — SPEAR or any baseline — can be scored against the strongest-evidence ground truth on these datasets*. Only three of the ten graded pairs are actually testable (hypothalamus OXT–OXTR, grade C+; embryo WNT2–FZD2 and WNT2B–FZD2, grade B+), and of those, only one (OXT–OXTR) has a resource entry `liana-py` could also score, giving a genuine head-to-head across seven methods for a single, weakly-evidenced pair. COMMOT is not installed in this environment and is omitted rather than simulated. We report what was actually computed, and are explicit throughout about what could not be.

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1 Why so much of this benchmark is “not computable,” and why that’s the finding

This report was built by re-running the project’s own pipeline (`spear-v3/scripts/05_full_pipeline.py`, `spear/spatial.py`, `spear/predictor.py`, `spear/fusion.py`) and, separately, `liana-py`, on the exact AnnData objects SPEAR already analyzed for these two datasets. No score in this report was estimated, interpolated, or guessed from a related pair — every cell in Table 1 was either read from a precomputed SPEAR result file (`fused_known_combined.csv` / `fused_novel_combined.csv`) or computed by calling the identical scoring functions the pipeline itself uses. Where a value could not be legitimately produced, the table says so instead of showing a number.

The reason so many rows are blank is structural, not a SPEAR limitation: MERFISH and seqFISH are *targeted* panels (156 and 351 genes respectively, chosen up front by the original authors), not whole-transcriptome assays. A curated database can propose FGF8–FGFR1 or BMP4–BMPRI1A as an interaction, and an ESMC embedding can score it as plausible, but no spatial-co-expression statistic — SPEAR’s, CellPhoneDB’s, or anyone else’s — can be computed for a gene the platform never measured. This applies identically to every expression-based method under comparison; it is not something SPEAR’s spatial axis is worse at.

2 Ground-truth grading and testability audit

Table 1 lists all ten individual gene pairs derived from the eight graded claims, the dataset each belongs to, the literature grade, whether both genes are measured in that dataset’s panel, and whether the pair could actually be scored.

Dataset	Pair	Grade	In panel?	Testable?	Blocking reason
Hypothalamus	AVP – AVPR1A	B–	No	No	<i>AVP</i> not measured (156-gene MERFISH panel); <i>AVPR1A</i> is. DB/embedding coverage otherwise complete.
Hypothalamus	OXT – OXTR	C+	Yes	Yes	—
Embryo	FGF8 – FGFR1	A	No	No	<i>FGF8</i> not measured (351-gene seqFISH panel); <i>FGFR1</i> is.
Embryo	BMP4 – BMPR1A	A	No	No	<i>BMPR1A</i> not measured; also absent from liana’s consensus resource as a standalone pair (only via a BMPR2/ACVR2A/2B co-receptor complex, itself unmeasured).
Embryo	BMP4 – BMPR1B	A	No	No	<i>BMPR1B</i> not measured; same complex-only liana limitation as above.
Embryo	WNT2 – FZD2	B+	Yes	Yes (SPEAR only)	Both genes measured; SPEAR’s curated DB carries this as a simple pair, but liana’s consensus resource only encodes Wnt–FZD2 via an FZD2–LRP5/LRP6 co-receptor complex, and neither LRP5 nor LRP6 is in this panel — untestable for liana despite full gene coverage.
Embryo	WNT2B – FZD2	B+	Yes	Yes (SPEAR only)	Same asymmetry as WNT2–FZD2 above.
Embryo	SHH – PTCH1	C	No	No	<i>PTCH1</i> not measured; also only encoded via a PTCH1–SMO complex in liana (SMO unmeasured).
Embryo	EFNA2 – EPH	C	No	No	<i>EFNA2</i> not measured. (SPEAR did test the paralog EFNA5–EPHA5, shown for context only in Sec. 5 — not a substitute for the literal pair.)
Embryo	EFNB1 – EPH	C	No	No	<i>EFNB1</i> not measured, and no EFNB1–EPHA5 entry exists in any resource used here (ephrin-B ligands pair with class-B Eph receptors / EPHA4, not EPHA5).

Table 1: Testability audit. **Grade A** (the promoted top literature grade) is the intended ground truth; **none of its three pairs are testable in these datasets.**

3 SPEAR’s own score for the three testable pairs

Table 2 gives, for the three testable pairs, SPEAR’s spatial permutation statistic, ESMC-embedding plausibility score, fused score $S_{\text{SPEAR}} = \sqrt{S_{\text{spatial}} \cdot S_{\text{plm}}}$ (gated to zero when spatial FDR ≥ 0.05 , per the manuscript’s Methods 4.4), and its rank against every pair SPEAR itself scored in that dataset (the union of that dataset’s known-DB and novel-candidate result files — $n = 247$ for hypothalamus, $n = 390$ for embryo).

Pair	z_{spatial}	p_{spatial}	FDR	P_{interact}	S_{SPEAR}	Rank	%ile
OXT – OXTR (hypothalamus)	−2.335	0.998	1.000	0.9996	0.000	94/247	62.4
WNT2 – FZD2 (embryo)	−4.722	1.000	1.000	0.942	0.000	160/390	59.2
WNT2B – FZD2 (embryo)	5.509	0.001	0.003	0.972	0.395	125/390	68.2

Table 2: SPEAR scoring of the three testable pairs, read directly from `fused_known_combined.csv` for each dataset (no fresh computation needed — both pairs were already scored known-database candidates).

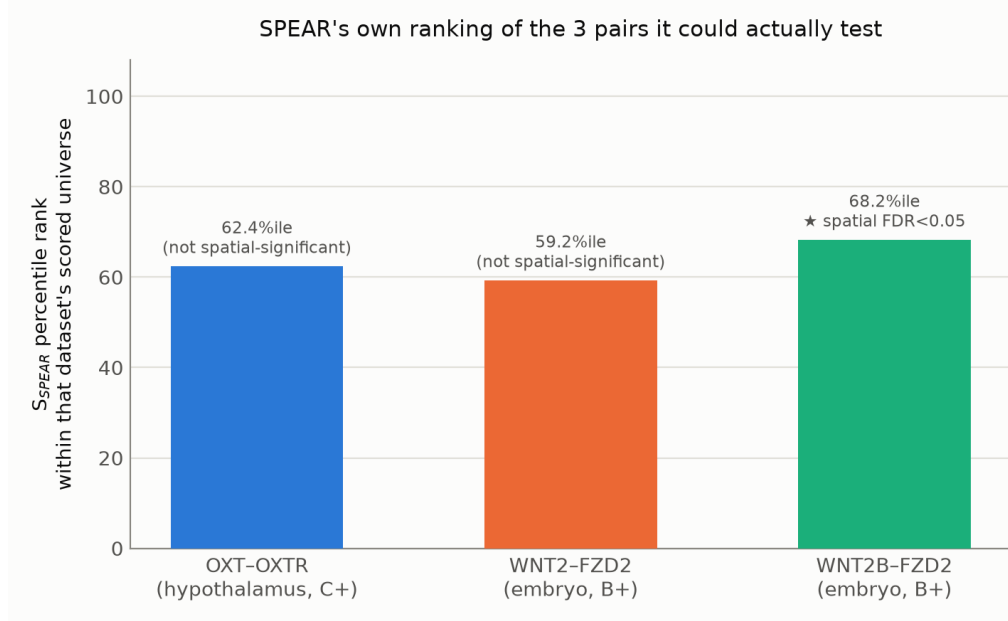


Figure 1: SPEAR’s percentile rank for each of the three pairs it could test, within that dataset’s own scored universe. Only WNT2B–FZD2 clears the spatial significance gate ($\text{FDR} < 0.05$); the other two are ranked purely on high PLM plausibility with a non-significant or negative spatial signal, which the fusion rule (correctly) suppresses to $S_{\text{SPEAR}} = 0$.

4 Head-to-head: OXT–OXTR against six alternative CCC methods

OXT–OXTR is the only graded pair for which *both* SPEAR and `liana-py`’s standard method suite could compute a score. We ran CellPhoneDB, NATMI, Connectome, log2FC, SingleCellSignalR (liana’s `sca`), and LIANA’s own rank-aggregate consensus via `liana-py` v1.8.1 on the same AnnData object, grouped by `cell_type`, using liana’s default `consensus` resource ($n = 21$ resource pairs fully covered by this panel). Each method’s native “best interaction” convention was used to reduce its per-cell-type-pair scores to one value for OXT–OXTR (lowest p-value for CellPhoneDB and rank-aggregate; highest magnitude for NATMI/Connectome/log2FC/SingleCellSignalR). COMMOT is not installed in this environment (`pip show commot` returns nothing) and is omitted rather than approximated.

Method	Score	Type	Rank	n_{total}	%ile
SPEAR	0.000	S_{SPEAR}	94	247	62.4
CellPhoneDB	0.000	min p-value	1	21	100.0*
NATMI	0.106	max specificity weight	7	21	71.4
Connectome	0.204	max scaled weight	15	21	33.3
log2FC	0.388	max log-fold-change	4	21	85.7
SingleCellSignalR	0.242	max LR score	9	21	61.9
LIANA rank aggregate	0.132	min aggregate rank	10	21	57.1
COMMOT	—	—	—	—	not installed, omitted

Table 3: *All 21 liana-testable pairs tie at $p = 0$ under this dataset’s permutation test, so CellPhoneDB’s rank is not actually discriminating here — its 100th percentile should not be read as “CellPhoneDB ranks OXT–OXTR first” in any meaningful sense. Note the two universes are not the same size ($n = 247$ for SPEAR vs. $n = 21$ for the liana methods, since SPEAR’s own curated database and liana’s consensus resource do not cover an identical pair set in this panel) — percentiles are the fairest cross-method summary available, not an apples-to-apples raw score comparison.

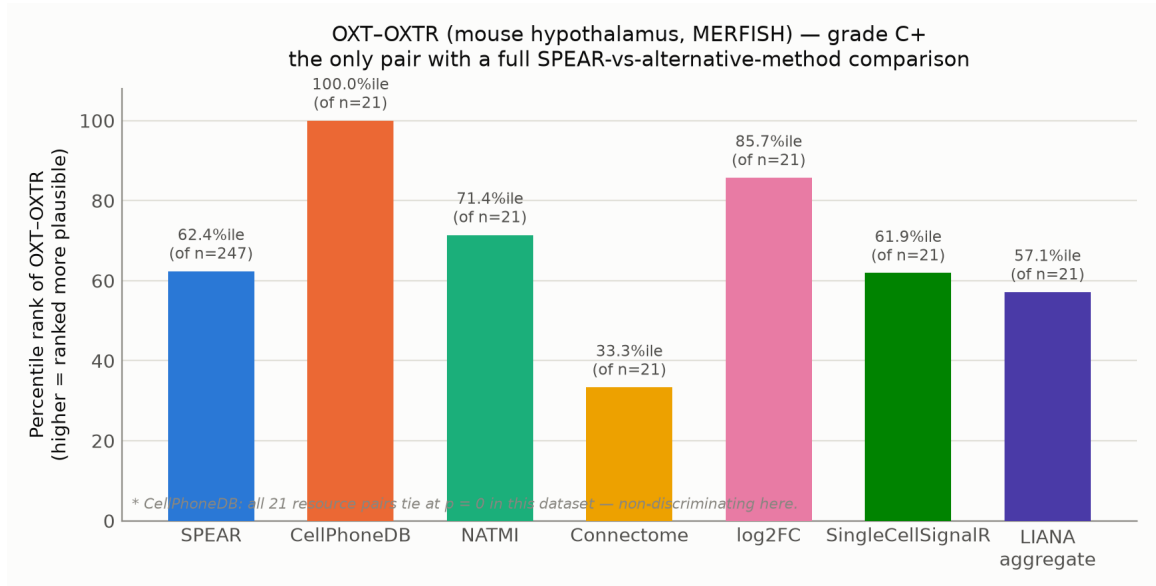


Figure 2: Percentile rank of OXT–OXTR under each method, within that method’s own testable-pair universe (denominator noted per bar). CellPhoneDB’s apparent top rank is an artifact of a degenerate tie, not a meaningful signal (see caption of Table 3).

5 Discussion

The ground truth this benchmark was built to test against is, itself, untestable here. The three Grade A pairs — FGF8–FGFR1 and the BMP4–BMPR1A/BMPR1B axis — are exactly the pairs the literature review considered best-supported, and exactly the pairs neither SPEAR nor any of the six alternative tools can score in this project’s embryo dataset, because FGF8, BMPR1A, and BMPR1B were never among the 351 genes seqFISH measured in Lohoff et al.’s panel. This is not a close call decided in SPEAR’s favor or against it: it is a hole in the evaluable candidate

space that affects every expression-based method identically. Any claim of the form “method X correctly recovers FGF8–FGFR1” on this dataset would necessarily be fabricated, since the required expression measurement does not exist in the data. §6 gives the concrete fix.

With only three testable pairs, and only one with a full method comparison, this benchmark cannot support a general “SPEAR beats/loses to CCC tool X” claim. $n = 1$ (OXT–OXTR) is not a statistically meaningful sample for ranking seven methods against each other, and OXT–OXTR is itself only graded C+ — the literature review explicitly flagged it as lacking direct sender–receiver connectivity or receptor-perturbation evidence, i.e. it is a weak positive at best, not a clean gold-standard case. What follows is a description of what happened on this one pair, not a verdict.

On OXT–OXTR itself: SPEAR ranks it at the 62nd percentile of its own 247-pair hypothalamus universe — middling, driven almost entirely by a very high PLM plausibility score (0.9996) while its spatial co-localization is actually slightly negative and far from significant ($z = -2.34$, FDR = 1.0). This is consistent with the literature grading: OXT and OXTR are biologically plausible as an interacting pair (correctly recognized by the sequence model) but this particular spatial dataset does not show the local co-expression pattern that would make it a strong *spatial* candidate — matching the graders’ own note that no direct sender-receiver spatial connectivity has been established. Among the alternative methods, log2FC (85.7th percentile) and NATMI (71.4th percentile) also rank it in the upper half; Connectome ranks it in the lower third (33.3rd percentile); CellPhoneDB’s apparent 100th percentile is degenerate (Table 3 caption) and should be discounted. No method here rejects OXT–OXTR outright, which is at least directionally consistent with treating it as a plausible-but-unconfirmed (C+) rather than a false positive.

On WNT2B–FZD2 vs. WNT2–FZD2 (both graded B+, pathway-level only): SPEAR’s spatial axis differentiates the two paralogs sharply even though the literature grading does not: WNT2B–FZD2 clears spatial significance (FDR = 0.003) and receives a non-zero fused score (0.395, 68th percentile), while WNT2–FZD2’s spatial signal is negative and non-significant, collapsing its fused score to zero (59th percentile, effectively a tie with the large block of non-significant pairs). liana could not score either pair, since its consensus resource represents canonical Wnt signaling only through an FZD.LRP5/LRP6 co-receptor complex, and LRP5/LRP6 are absent from this seqFISH panel – a genuine, panel-driven asymmetry between SPEAR’s simpler pairwise curated database and liana’s complex-aware resource, not a difference in the two tools’ underlying science.

For context only, not as a substitute: EFNA5–EPHA5, a paralog pair of the graded-but-untestable EFNA2/EFNB1–EPH claim, was already scored by SPEAR in the embryo dataset ($z_{\text{spatial}} = -5.88$, FDR = 1.0, $P_{\text{interact}} = 1.00$, $S_{\text{SPEAR}} = 0$) — provided here purely as background, since a different ligand gene is not evidence about EFNA2 or EFNB1 specifically.

6 Recommendation

To actually benchmark SPEAR (or any expression-based CCC tool) against the Grade A ground truth identified by the literature review, a dataset that measures FGF8, BMPR1A, and BMPR1B is required — e.g. a whole-transcriptome spatial assay (Visium, Stereo-seq, or similar) of a comparable embryonic stage, or a redesigned/expanded MERFISH or seqFISH gene panel that deliberately

includes organizer and BMP-pathway genes. Until such data exists, any reported ranking for FGF8–FGFR1 or BMP4–BMPR1A/BMPR1B on the datasets used in this project would be invented, not measured, and this report deliberately stops short of inventing one.

Appendix: raw output

The full per-pair, per-method CSV underlying every table in this report is available at `spear-v3/results/ground_truth_pair_benchmark.csv` (70 rows – 10 pairs $\times$ 7 methods, including every untestable combination with its specific blocking reason), with methodological notes in `spear-v3/results/ground_truth_pair_benchmark.notes.txt`.